rainmaker

_travelers guide _for creating microclimates

16	64	128
delay taps	comb taps	presets

› then noon = 12 o'clock tags = the panel drawing

stereo 16-tap rhythm delay + 64-tap comb resonator · every tap has its own filter, pan and pitch · 128 presets

01 what it is

rm_01

two effects	a 16-tap stereo rhythm delay + a 64-tap comb resonator, 0.1 ms to 20 s
every tap	its own level, pan, filter (lp · bp · hp) and pitch shift
the comb	long = echoes · short + high feedback = karplus-strong plucks and pitched ringing
inside	96 kHz / 32-bit fpga · 128 preset slots

02 two ways to edit

rm_02 · the big idea

`[ edit/tap# ]`	the white button top left · flips the 16 blue buttons between two jobs
› edit	each blue button opens a page: rand · grid · piles · time · div · wave · level · rate · qty · size · tap# · slip · pitch · type · amt · reverse
› tap#	the same 16 buttons pick taps 1 – 16
then	a white tap edit button (mute · level · pan · cut · q · type · pitch shift) + the encoder
lost?	press `[ edit/tap# ]` again to swap modes

10 first flight

rm_10 · a clean start

1. `[ load ]` › preset 1 (1-tap) › push
2. sound into `in L` · `out L` `out R` out
3. `mix` to noon · tap `[ tempo ]` three times
4. raise delay feedback for repeats
5. `[ edit/tap# ]` to taps › `[ mute ]` › light more taps
6. save it before experimenting

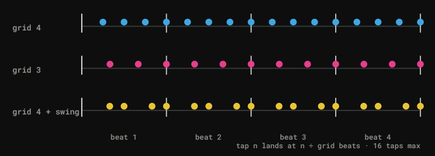
03 the taps

rm_03 · tap edit buttons

`[ mute ]`	in this view the 16 buttons toggle each tap
`[ level ]`	0 – 15, exponential · 0 = off
`[ pan ]`	7L … 0 … 7R (balance in stereo)
`[ cut ]` `[ q ]` `[ type ]`	filter per tap: cutoff ~Hz, q 0 – 15, none · lp · bp · hp
`[ pitch ]` twice	–16 … +15 semitones · press again for detune
all at once	in edit mode: push the encoder, turn = every tap · push twice = 8 ramp shapes

_fig.01 : where the taps land

tap grid · groove



04 rhythm · time

rm_04 · fig.01

`[ grid ]`	taps per beat: 1 2 3 4 6 8 12 16 · tap n lands at n ÷ grid beats
`[ piles ]`	the tap stack · taps share the time of the last tap in the pile
`[ tempo ]`	tap 3 times · a `clk` input overrides it · clock div /1 – /256
`[ type ]` `[ amt ]`	groove: 16 types (swing, accelerando, waltz, random, reflections …) · amount 0 – 100
`[ tap# ]` (feedback)	which tap repeats: 1 – 16, all, or the comb size · push = ping-pong

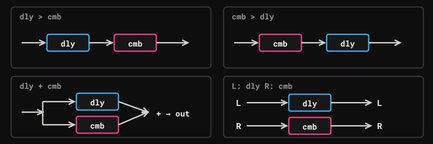
05 the comb

rm_05 · comb pages

`size` pot	the comb length · its pitch when short
`[ taps+fb ]`	1 – 64 taps · then the feedback flavour: guitar · sitar · clarinet · raw
`[ patt+slope ]`	16 tap patterns (fibonacci, early, late …) · then the level slope
`[ clk src ]`	what sets the size: comb clk · size cv · encoder · the delay time
`1v/oct`	play it like an oscillator · try preset 38, karplus-strong

_fig.02 : routing

routing+config, first press



06 routing · trigger · mod

rm_06 · fig.02

`[ routing+config ]` press to cycle: routing · input mode · trigger mode

routing	dly › cmb · cmb › dly · both in parallel · L = dly, R = cmb
trigger	big red button + `trig` jack: ping, freeze, reverse, randomize, clear, next preset, mutes …
`mod a` `mod b`	press `[ mod a ]` or `[ mod b ]`, turn to a destination, push to lock it
note	mod A is normalled to mod B · only mod B can set delay time

07 quiet? distorted?

rm_07 · levels

`[ meters+gain ]`	shows in · out · delay and comb feedback · sets their output gain
delay gain	about 8 – 10 for 1 – 2 taps · lower it as taps pile up
inputs	attenuators near noon suit normal eurorack levels · keep IN off the top
high q	tap filters at high q can distort · turn the input down
two mixes	`mix` pot = analog dry/wet · the wet/dry page = inside the delay

08 presets · patches

rm_08 · manual

save — save › turn › push
press save again to name it (10 chars) · "saved!"

load — load › turn › push
hold load as you push to load only the active effect · knobs take over once moved

plucked strings — preset 38
karplus-strong: a trigger into trig, pitch cv into 1v/oct

worth a look — 66 · 68 · 82 +
66 modal ring · 68 delay*size · 82 – 128 by richard devine

_fig.03 : the panel

numbers match the legend



`edit` `edit/tap#` · pages ↔ per-tap
`pages / taps` `16 blue buttons` · edit pages · or taps 1 – 16
`tap edit` `white row` · mute · level · pan · cut · q · type · pitch
`enc` `encoder` · turn · push · push+turn = coarse
`pots` `8 large` · feedback · pitch · tone · size · comb fb · mix · mod a · mod b
`cv` `cv jacks` · 0 – 10 V · small knobs scale them
`trig` `trigger` · big red button + jack
`tempo` `tempo` · tap 3× · clk in overrides
`presets` `red column` · routing · meters · save · load
`mod` `mod a` · `b` · button = pick destination

09 specs

rm_09

audio	±5 V in and out · in L feeds in R
cv · clock	cv 0 – 10 V · clk in / out 0 – 5 V
delay	0.1 ms – 20 s · 96 kHz / 32-bit
size	36 hp · 44 mm deep
power	270 mA +12 V (nearly 300 at start) · 24 mA –12 V